

Caffeine Consumption and the Risk of Primary Open - Angle Glaucoma: A Prospective Cohort Study

Purpose We investigated whether caffeine, which transiently increases intraocular pressure (IOP) is associated with risk of primary open-angle glaucoma (POAG). **Methods** We followed 79,120 women from 1980 and 42,052 men from 1986 to 2004 who were 40+ years old, did not have POAG, and reported receiving eye examinations. Information on caffeine consumption, potential confounders and POAG diagnoses were repeatedly updated in validated follow-up questionnaires. We confirmed 1,011 incident POAG cases with medical record review. Cohort-specific and pooled analyses across cohorts were conducted to calculate multivariable rate ratios (RR). **Results** Compared with daily intake of < 150 mg, the pooled multivariable RRs were 1.05 [95% Confidence Interval (CI), 0.89–1.25] for consuming 150–299 mg, 1.19 [95% CI, 0.99–1.43] for 300 – 449 mg/day, 1.13 [95% CI, 0.89–1.43] for 450–559 mg and 1.17 [95% CI, 0.90, 1.53] for 600+ mg+ [p for trend = 0.11]. However, for consuming 5+ cups of caffeinated coffee daily, RR was 1.61 [95% CI, 1.00, 2.59; p for trend=0.02]; tea or caffeinated cola intake were not associated with risk. Greater caffeine intake was more adversely associated with POAG among those reporting family history of glaucoma, particularly in relation to POAG with elevated IOP (p for trend =0.0009; p-interaction=0.04). **Conclusion** Overall caffeine intake was not associated with increased risk of POAG. However, in secondary analyses, caffeine appeared to elevate risk of high-tension POAG among those with a family history of glaucoma; this may be due to chance, but warrants further study.